

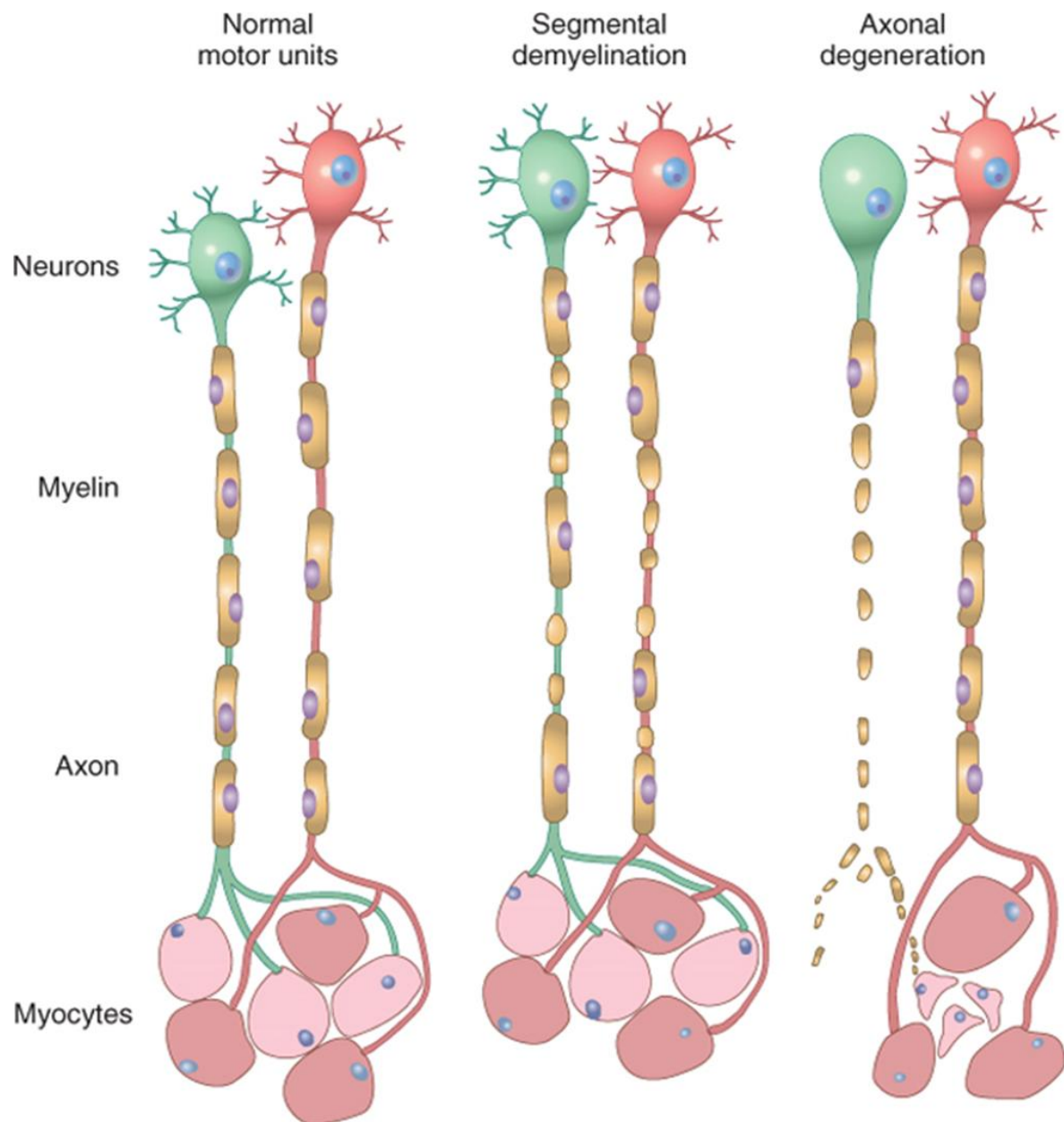
SKELETAL MUSCLE

- Normal skeletal muscle development and function critically depend on a tight integration with the central and peripheral nervous systems .
- The peripheral nerves and skeletal muscles permit purposeful movement and provide the brain with sensory information about our surroundings.

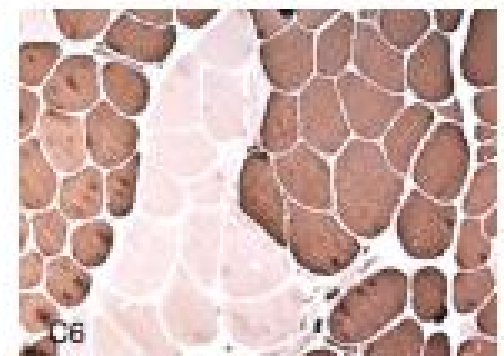
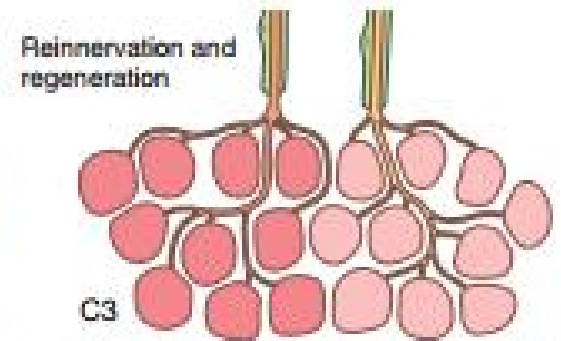
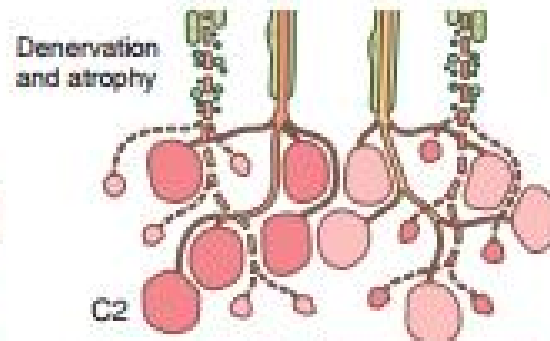
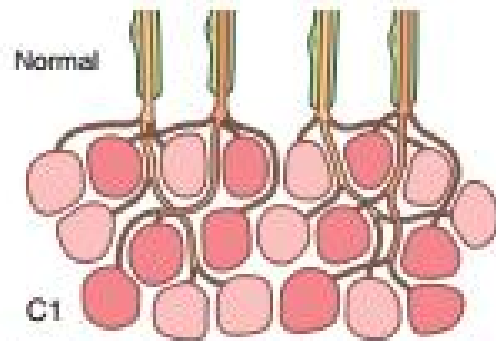
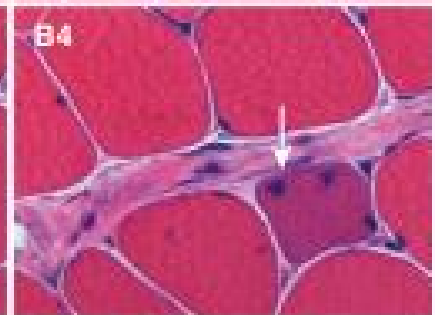
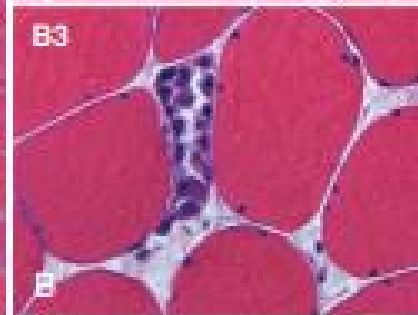
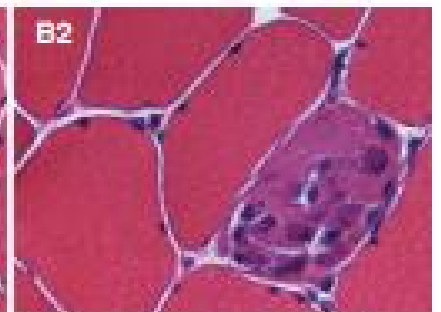
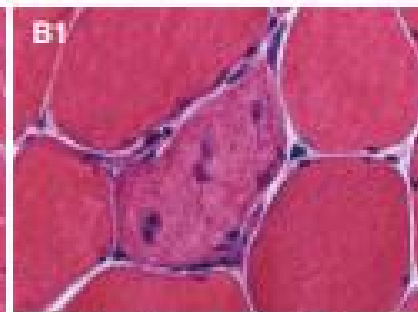
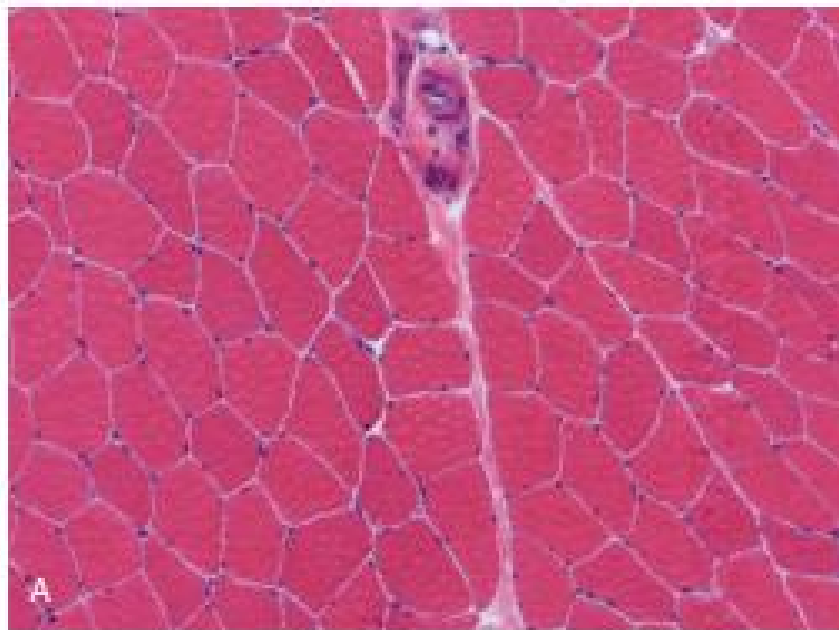
- The principal element of this integrated system is called the **motor unit**.
 - It is comprised of
 - A motor neuron in the brain or spinal cord
 - Its associated peripheral axon
 - Distal neuromuscular junction and
 - The skeletal muscle fibers that are innervated.

- Depending on the nature of the nerve fiber doing the enervation, the associated skeletal muscle develops into one of two major subpopulations;
 - **type I** "slow twitch" or
 - **type II** "fast twitch" fibers.
- The different fibers are distinguishable by unique metabolic and biochemical attributes, and can be identified using specific staining techniques.

- A single "**type I**" or "**type II**" neuron will innervate multiple muscle fibers and these fibers are usually randomly scattered in a "**checkerboard pattern**" within a circumscribed area within the larger muscle.



	Type I	Type II
Action	Sustained force	Fast movement
Activity type	Aerobic exercise	Anaerobic exercise
Power produced	Low	High
Resistance to fatigue	High	Low
Lipid content	High	Low
Glycogen content	Low	High
Energy metabolism	Low glycolytic capacity, high oxidative capacity	High glycolytic capacity, low oxidative capacity
Mitochondrial density	High	Low
Myosin heavy chain gene expressed	<i>MYH7</i>	<i>MYH2, MYH4, MYH1</i>
Color	Red (high myoglobin content)	Pale red / tan (low myoglobin content)



- Skeletal muscle disease can be divided into
 - ❑ Disorders characterized by neurogenic changes or myofiber atrophy
 - ❑ The more common muscular dystrophies
 - ❑ Selected congenital, toxic, and inflammatory myopathies
 - ❑ Disorders of the neuromuscular junction.
 - ❑ Primary skeletal muscle tumors.

MUSCLE ATROPHY

- Muscle atrophy is a **non-specific response** in a variety of muscle disorders.
- It is characterized by **abnormally small myofibers**; the type of fibers affected by the atrophy, their distribution in the muscle, and their specific morphology help identify the etiology of the atrophic changes.

- Clearly loss of muscle innervation causes atrophy of the associated fibers.
- **Neurogenic atrophy** is characterized by involvement of *both fiber types* and by clustering of myofibers into small groups.
- **Simple disuse** (e.g., Prolonged bed rest, immobilization to allow healing of a bone fracture, etc.) Can also cause profound atrophy.

- **Exogenous glucocorticoids or endogenous hypercortisolism** (e.g., in Cushing syndrome) are another cause of muscle atrophy, typically involving proximal muscle groups more than distal ones.
- Disuse- and steroid-induced atrophy primarily affects the **type II fibers** and causes a random distribution of the atrophic myofibers.

- Finally, atrophic myofibers are also found in **myopathies**.
 - The finding of additional morphologic changes like
 - myofiber degeneration and regeneration
 - chronic remodeling of the tissue
 - inflammatory infiltrates
 - are features that suggest a myopathic etiology.

Neurogenic Atrophy

- Deprived of their normal enervation, skeletal fibers undergo progressive atrophy.
- It is important to recall that loss of a **single neuron** will affect all muscle fibers in a motor unit, so that the atrophy tends to be **scattered over the field**.
- However, following re-enervation, adjacent intact neurons send out sprouts to engage the neuromuscular junction of the previously de-enervated fibers.

- Once the new connection is established these fibers assume the type of the innervating neuron.
- In this manner, whole groups of fibers can eventually fall under the influence of the same neuron, and become the same fiber type (**fiber type grouping**).
- In that setting, if the relevant enervating neuron now becomes injured, rather **large coalescent groups of fibers** are cut off from the trophic stimulation and wither away (**grouped atrophy**,) a hallmark of **recurrent neurogenic atrophy**.

MUSCULAR DYSTROPHY

- The muscular dystrophies are a heterogeneous group of inherited disorders, often presenting in childhood, that are characterized by **progressive degeneration of muscle fibers leading to muscle weakness and wasting.**
- Histologically, in advanced cases muscle fibers are replaced by **fibrofatty tissue.**
 - This histologic feature distinguishes dystrophies from myopathies, which also present with muscle weakness.

X-Linked Muscular Dystrophy (Duchenne and Becker Muscular Dystrophy)

- The two most common forms of muscular dystrophy are X-linked:
 - Duchenne muscular dystrophy (DMD)
 - Becker muscular dystrophy (BMD).

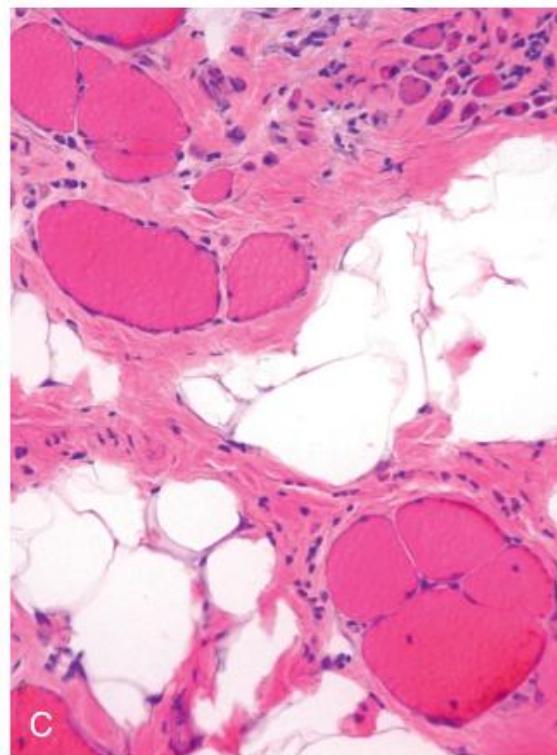
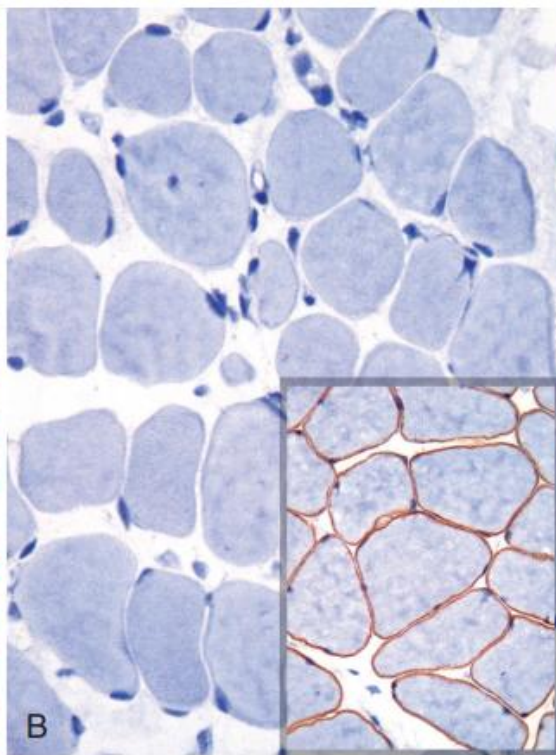
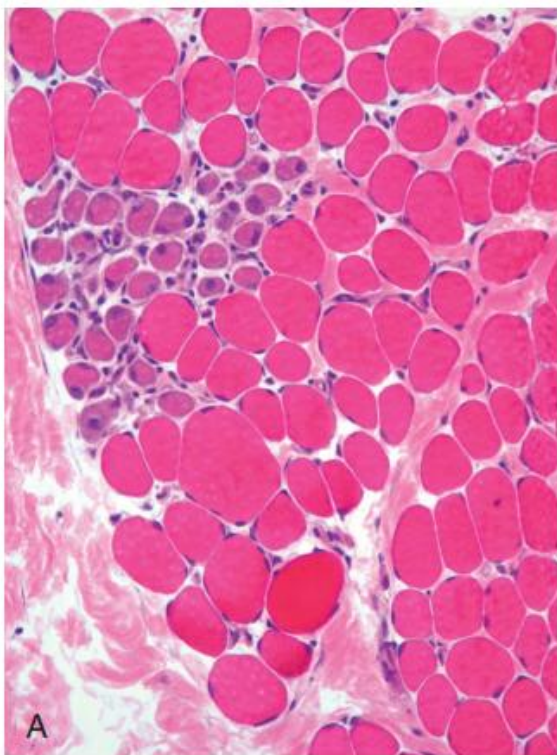
- **DMD** is the **most severe** and the **most common form** of muscular dystrophy, with an incidence of about 1 per 3500 live male births.
- DMD becomes clinically evident by age 5, with progressive weakness leading to wheelchair dependence by age 10 to 12 years, and death by the early 20s.
- Although the **same gene** is involved in both BMD and DMD, **BMD** is **less common** and much **less severe**.

Morphology

- The histologic features of DMD and BMD are similar and include **marked variation in muscle fiber size**, caused by concomitant myofiber hypertrophy and atrophy.
- Many of the residual muscle fibers show a **range of degenerative changes**, including
 - fiber splitting
 - necrosis
- whereas other fibers show evidence of regeneration, including
 - sarcoplasmic basophilia,
 - nuclear enlargement, and
 - nucleolar prominence.
- Connective tissue is increased throughout the muscle.

...cont.

- The definitive diagnosis is based on the
 - demonstration of abnormal staining for dystrophin in immunohistochemical preparations or
 - by western blot analysis of skeletal muscle.
- In the late stages of the disease,
 - extensive fiber loss and
 - adipose tissue infiltration are present in most muscle groups.
- Changes in cardiac muscle in either DMD or BMD include
 - variable degrees of fiber hypertrophy and
 - interstitial fibrosis.



pathogenesis

- DMD and BMD are caused by abnormalities in the **dystrophin gene** located on the short arm of the X chromosome (Xp21).
- Dystrophin is a large protein (427 kD) that is expressed in a wide variety of tissues, including muscles of all types, brain, and peripheral nerves.
- Dystrophin attaches portions of the **sarcomere to the cell membrane**, maintaining the structural and functional integrity of skeletal and cardiac myocytes.

- The role of dystrophin in transferring the force of contraction to connective tissue has been proposed as the basis for the myocyte degeneration that occurs with dystrophin defects, or with changes in other proteins that interact with dystrophin.
- Muscle biopsy specimens from individuals with DMD show virtually no dystrophin by either immunohistochemical staining or western blot analysis, explaining the **greater severity** of their presentation.
- In comparison, individuals with BMD show diminished amounts of an abnormal molecular weight dystrophin, reflecting mutations that apparently permit limited synthesis of a defective (but still partially active) protein.

- Approximately **two-thirds** of the cases are **familial**, with the remainder representing new mutations.
- In affected families, **females are carriers**; they are clinically **asymptomatic** but often have **elevated serum creatine kinase** and can show **mild histologic abnormalities** on muscle biopsy.
- **Female carriers**, however, are **at risk for developing dilated cardiomyopathy**.

Clinical Features

- Boys with DMD are **normal at birth**, and **early motor milestones are met on time**.
- **Walking, however, is often delayed**, and the **first indications of muscle weakness are clumsiness and inability to keep up with peers**.
- Weakness begins in the **pelvic girdle muscles** and then extends to the **shoulder girdle**.

- **Enlargement of the calf muscles** associated with weakness, a phenomenon termed **pseudohypertrophy**, is an important clinical finding.
 - The increased muscle bulk is caused initially by an
 - increase in the size of the muscle fibers and then
 - as the muscle atrophies, by an increase in fat and connective tissue.

- Pathologic changes are also found in the heart, and patients may develop heart failure or arrhythmias.
- Although there are no well-established structural abnormalities of the central nervous system, cognitive impairment seems to be a component of the disease and is severe enough in some patients to be considered **mental retardation**.

- Serum creatine kinase is elevated during the first decade of life but returns to normal in the later stages of the disease, as muscle mass decreases.
- **Death** results from
 - **respiratory insufficiency**
 - **pulmonary infection**
 - **cardiac decompensation.**

- Boys with BMD develop symptoms at a later age than those with DMD.
 - The onset occurs in later childhood or in adolescence, and it is accompanied by a generally slower and more variable rate of progression.
 - Although cardiac disease is frequently seen in these patients, many have a nearly normal life span.

Autosomal Muscular Dystrophies

- Other forms of muscular dystrophy share many features of DMD and BMD but have distinct clinical and pathologic characteristics.
- Some of these muscular dystrophies affect **specific muscle groups**, and the formal diagnosis is based largely on the clinical pattern of muscle weakness.
- Several autosomal muscular dystrophies affect the proximal musculature of the trunk and limbs (similar to the X-linked muscular dystrophies), and are termed **limb girdle muscular dystrophies**.

- **Limb girdle muscular dystrophies** can be inherited either as autosomal dominant or autosomal recessive disorders; six dominant subtypes and ten recessive subtypes have been identified.
 - Mutations of the **sarcoglycan complex of proteins** cause four of the recessive forms of these dystrophies, with other forms being associated with other cytoskeletal proteins or **caveolin**.

Myotonic Dystrophy

- **Myotonia** is a sustained involuntary contraction of a group of muscles; it is the cardinal neuromuscular symptom in myotonic dystrophy.
 - Patients often complain of "stiffness" and have difficulty in releasing their grip, for instance, after a handshake.
 - Myotonia can often be elicited by percussion of the thenar eminence.

- Myotonic dystrophy is inherited as an **autosomal dominant trait**; it is associated with a **CTG trinucleotide repeat expansion** on chromosome 19 that affects the mRNA for the **dystrophila myotonia-protein kinase**.
- In normal subjects, fewer than 30 CTG repeats are present; disease develops with expansion of this repeat, and in **severely affected individuals several thousand repeats may be present**.

- Expansion of the trinucleotide repeat influences the eventual concentration of the protein product.
 - Like other "trinucleotide repeat disorders" (e.g., fragile X syndrome, Huntington disease), myotonic dystrophy tends to increase in severity and appear at a younger age in succeeding generations, a phenomenon termed **anticipation**.
- With each new generation there is expansion of the CTG repeats during gametogenesis, and this seems to correspond to the clinical feature of anticipation.

- The disease often presents in late childhood with gait abnormalities attributable to
 - weakness of foot dorsiflexors
 - it progresses to weakness of the intrinsic muscles of the hands and wrist extensors;
 - atrophy of facial muscles with ptosis ensues.

MYOPATHY

- The term **myopathy** encompasses a heterogeneous group of disorders, both morphologically and clinically.
- A diverse group of acquired disorders may manifest with muscle weakness, muscle cramping, or muscle pain.
- In most instances these are disorders of adults with acute or subacute onsets

Inflammatory Myopathies

- Inflammatory myopathies make up a heterogeneous group of rare disorders **characterized by immune-mediated muscle injury and inflammation.**
- Based on the clinical, morphologic, and immunologic features, three disorders-
 - polymyositis
 - dermatomyositis
 - inclusion body myositis
- These may occur alone or in conjunction with other autoimmune diseases, such as **Systemic Sclerosis.**
- ***Dermatomyositis*** is the most common inflammatory myopathy in children, in whom it appears as an isolated entity.
- In adults, it often manifests as a paraneoplastic disorder.

- Clinically, these diseases are characterized by **usually symmetric muscle weakness** initially affecting large muscles of the trunk, neck, and limbs.
 - Thus, tasks such as getting up from a chair or climbing steps become increasingly difficult.
 - In dermatomyositis an associated rash (classically described as a lilac or heliotrope discoloration) affects the **upper eyelids** and causes **periorbital edema**.

- Histologically, there is **infiltration by lymphocytes**, and both **degenerating and regenerating muscle fibers** are seen.
- The pattern of muscle injury and the location of the inflammatory infiltrates are fairly distinctive for each subtype.

- The immunologic evidence supports antibody-mediated tissue injury in dermatomyositis, whereas **polymyositis and inclusion body myositis seem to be mediated by CTLs.**
- ANAs are present in most patients.
 - Of these, only Jo-1 antibodies, directed against transfer RNA synthetase, are specific for this group of disorders .
- The diagnosis of these myopathies is based on
 - clinical features
 - laboratory evidence of muscle injury (e.g., increased blood levels of creatine kinase)
 - electromyography
 - biopsy.

1. Polymyositis

a. Clinical presentation

- i. Adults**
- ii. Bilateral proximal muscle weakness**

b. Micro

- i. Endomysial lymphocytic inflammation (mostly cytotoxic T8)**
- ii. Skeletal muscle fiber degeneration and regeneration**

2. Dermatomyositis

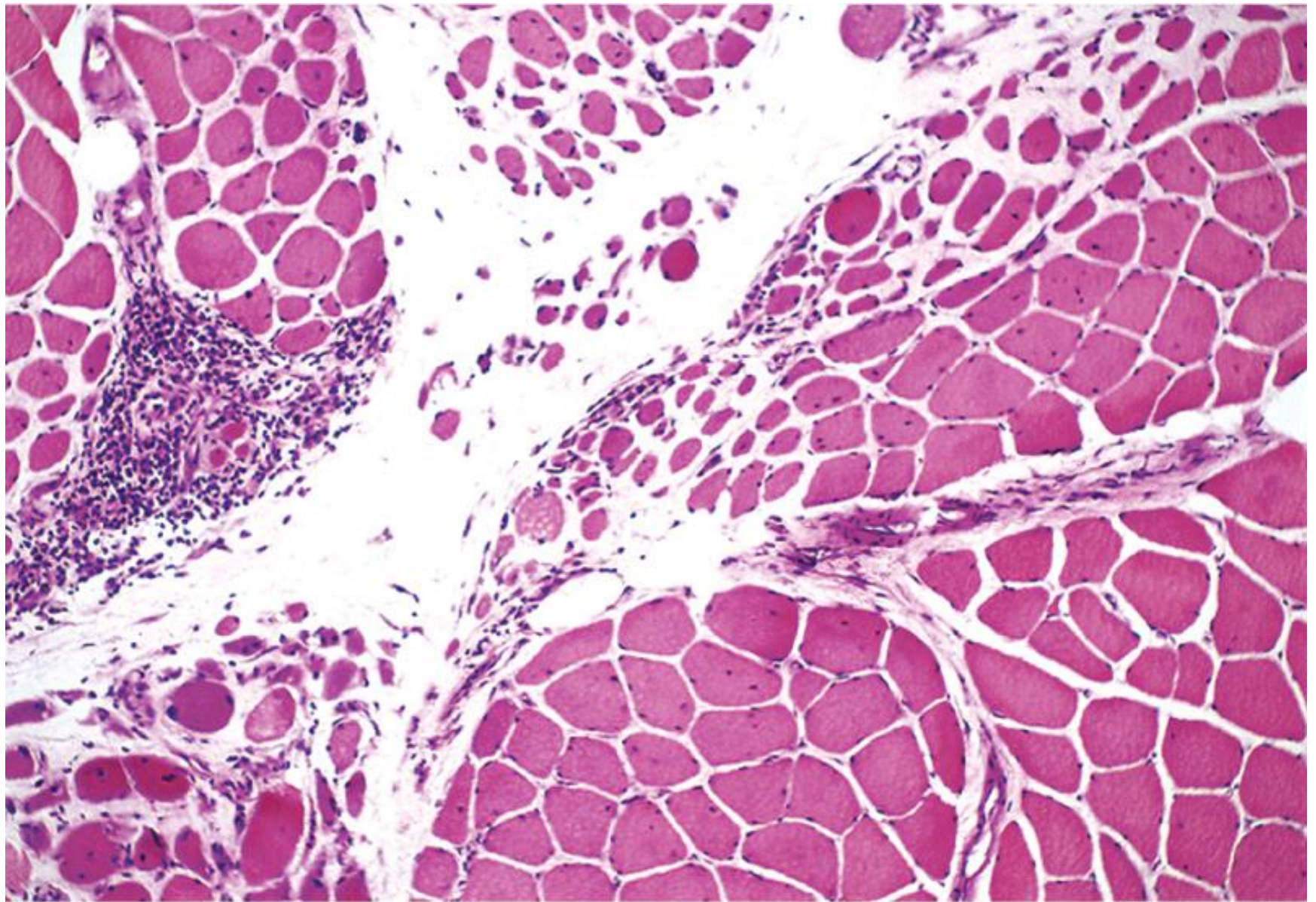
a. Clinical presentation

- i. Children or adults
- ii. Bilateral proximal muscle weakness
- iii. Skin rash of the upper eyelids
- iv. Periorbital edema

b. Micro

- i. Perimysial and vascular lymphocytic inflammation
- ii. *Perifascicular fiber atrophy*
- iii. Skeletal muscle fiber degeneration and regeneration

c. Increased risk of lung, stomach, and ovarian cancers



3. Inclusion body myositis

a. Clinical presentation

i. Adults > age 50

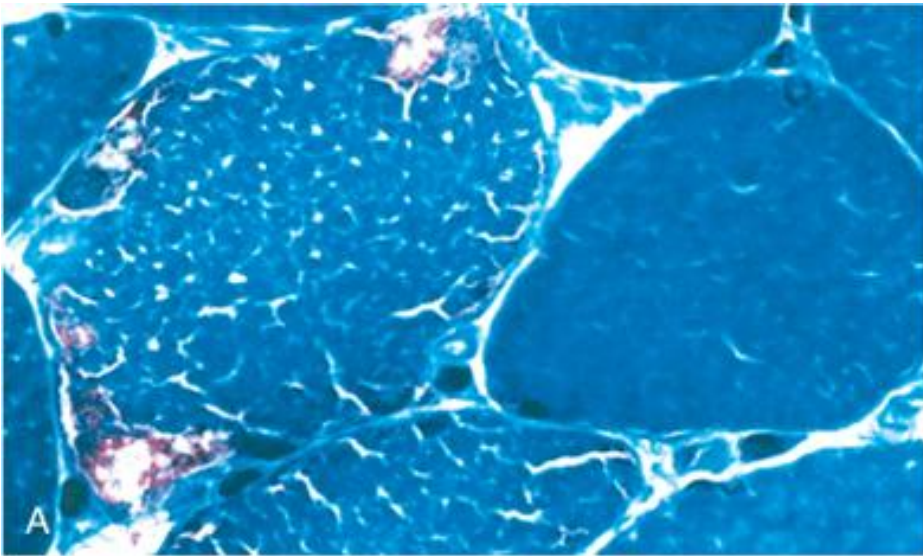
ii. Asymmetrical distal muscle weakness

b. Micro: cytoplasmic vacuoles with basophilic granules and amyloid

c. EM: filamentous inclusions

Congenital Myopathies

- Important subcategories include disorders caused by
 - inherited mutations of ion channels (channelopathies),
 - inborn errors of metabolism (exemplified by glycogen and lipid storage diseases)
 - mitochondrial abnormalities.



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Mitochondrial myopathy showing an irregular fiber with subsarcolemmal collections of mitochondria that stain red with the modified Gomori trichrome stain (ragged red fiber). B, Electron micrograph of mitochondria from biopsy specimen in A showing "parking lot" inclusions (arrowheads).

Toxic Myopathies

- Important subcategories include disorders caused by
 - intrinsic exposures (e.g. thyroxine) versus
 - extrinsic exposures (e.g., alcohol, therapeutic drugs).

- **Thyrotoxic myopathy** can present as either acute or chronic proximal muscle weakness, and can precede the onset of other signs of thyroid dysfunction.
 - Findings include myofiber necrosis, regeneration, and interstitial lymphocytes.

- **Ethanol myopathy** can occur with binge drinking, where there is an acute toxic rhabdomyolysis with accompanying myoglobinuria that can cause renal failure.
 - Clinically, the patient may acutely develop pain that is either generalized or confined to a single muscle group.
 - On histology, there is
 - myocyte swelling and necrosis
 - myophagocytosis, and
 - regeneration.

DISEASES OF THE NEUROMUSCULAR JUNCTION

Myasthenia Gravis

- Myasthenia gravis is an autoimmune disease with fluctuating muscle weakness that is caused by autoantibodies that target the neuromuscular junction
- The disease affects roughly 3 in 100,000 persons; it can present **at any age and has a predilection for women.**
- **Thymic hyperplasia** is found in **65%** and a **thymoma** in **15%** of patients.

- Circulating **antibodies** to the skeletal muscle acetylcholine receptors (**AChRs**) are present in nearly all patients, associated with a **decrease** in the number of AChRs.
- The disease can be transferred to animals with serum from affected patients, demonstrating the causal role of the anti-AChR antibodies.

...Pathogenesis

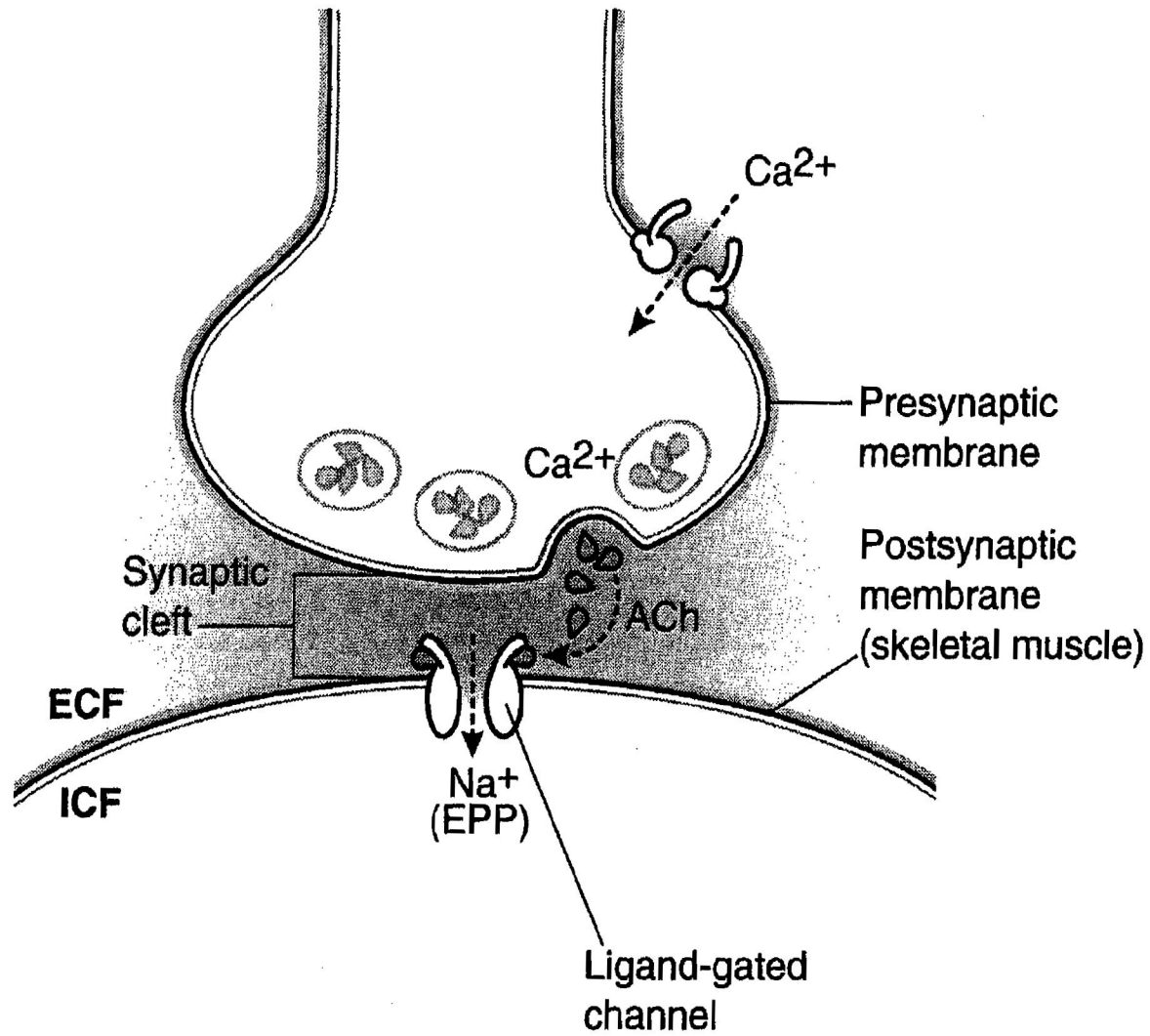
- In most cases, the autoantibodies against the AChR lead to loss of functional AChRs at the neuromuscular junction either by
 - (1) increasing the internalization and degradation of the receptors, and/or
 - (2) blocking the binding of acetylcholine (ACh) to its receptor.
- Notably, the autoantibodies do not appear to cause disease by inducing muscle destruction.

- Despite the evidence that anti-AChR antibodies function critically in the pathogenesis of the disease, antibody levels and neurologic deficit are not always correlated.
- The link between autoimmunity to AChRs and the thymic abnormalities is also unclear.
- Nevertheless, most patients show improvement after **thymectomy**.

... Clinical Features

- Typically, weakness is first noticed in the **extraocular muscles**; drooping eyelids (ptosis) and double vision (diplopia) cause the patient to seek medical attention.
- The generalized muscle weakness can fluctuate dramatically, with alterations occurring over the course of days, hours, or even minutes.

- Repetitive electrophysiologic stimulation typically elicits diminishing muscle strength, and patients show **marked improvement after administration of anticholinesterase agents**-the latter presumably by increasing the levels of ACh in the neuromuscular synapse; both maneuvers are diagnostically useful.



- **Sensory and autonomic functions are not affected** in myasthenia gravis.
- Respiratory compromise was a major cause of mortality in the past
- 95% of patients now survive more than 5 years after diagnosis because of improved treatment (anticholinesterase drugs, prednisone, plasmapheresis, and thymic resection), as well as ventilatory support.



Patient with myasthenia gravis showing ptosis of the left eye (A) followed by opening of the eye (B) after intravenous injection of Tensilon.



Lambert-Eaton Myasthenic Syndrome

- The Lambert-Eaton myasthenic syndrome characteristically develops as a **paraneoplastic** process, most commonly in the setting **of small-cell lung carcinoma (60% of cases)**; it can also occur in the absence of malignancy.

- Although individuals with Lambert-Eaton syndrome also present with muscle weakness, the syndrome is distinct from myasthenia gravis in several ways:
 - (1) anticholinesterase administration does not improve symptoms
 - (2) autonomic function is affected
 - (3) electrophysiologic studies demonstrate that repeated stimulation elicits increasing muscle strength.

- In these patients, the content of *ACh* is *normal* in neuromuscular junction synaptic vesicles, and the postsynaptic membrane is normally responsive to ACh, but fewer vesicles than normal are released in response to each presynaptic action potential.
- This is attributed to **antibodies that recognize presynaptic calcium channels**; a similar disease can be transferred to animals with these antibodies.

SKELETAL MUSCLE TUMORS

- Skeletal muscle neoplasms are **almost all malignant**.
- The benign rhabdomyoma is rare.
- Cardiac rhabdomyomas are examples of hamartomas.

- Extracardiac rhabdomyomas take several forms, may affect adults or children and are very rare.
 - They are clinically **benign** and usually have no great biologic significance once they have been accurately diagnosed.

Adult Rhabdomyoma

- Adult rhabdomyoma (A-RM) is a rare
- benign mesenchymal tumour with mature skeletal muscle differentiation and a predilection for the **head and neck** region.

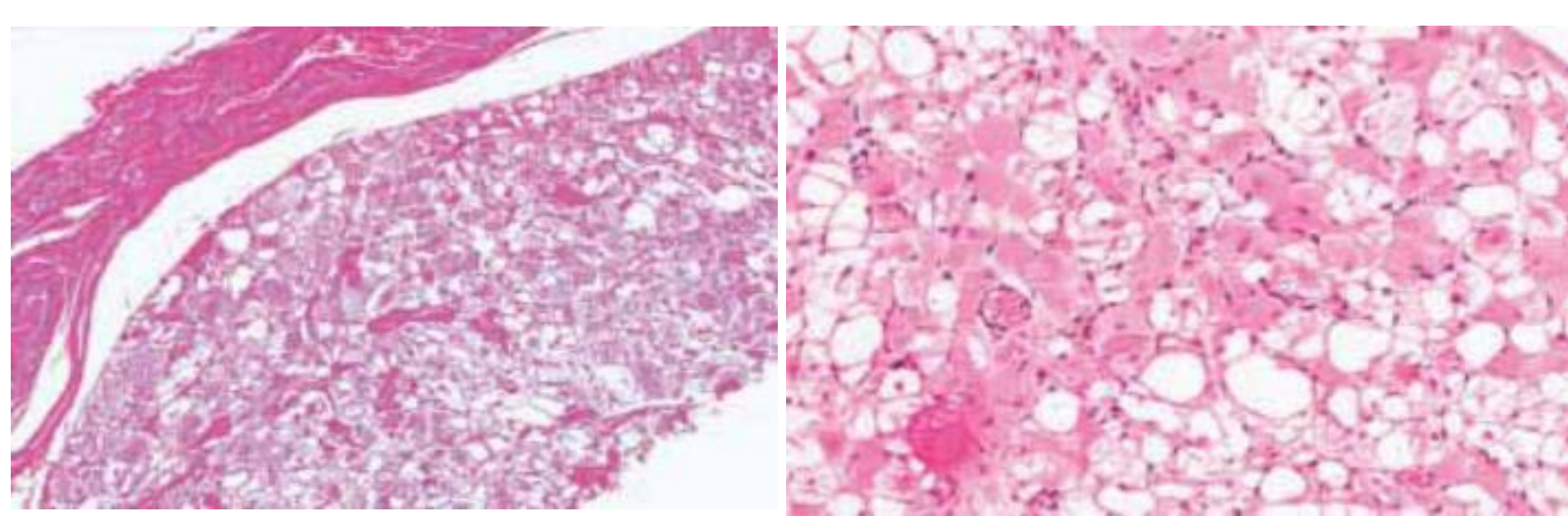
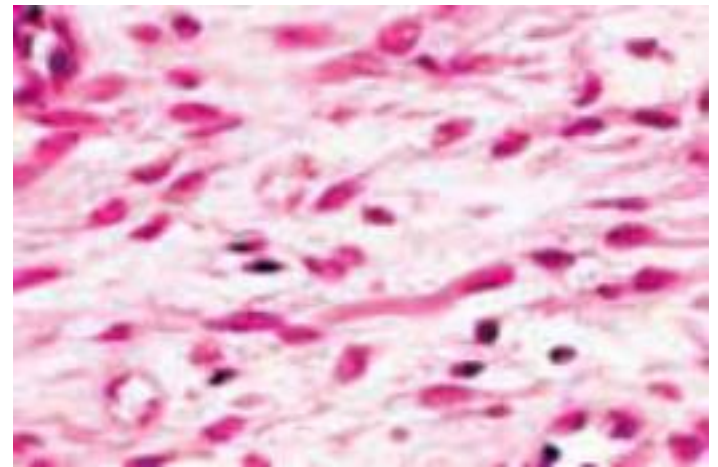
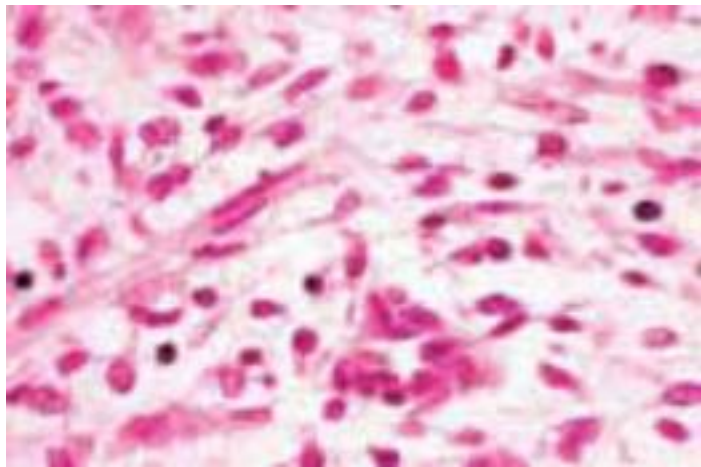
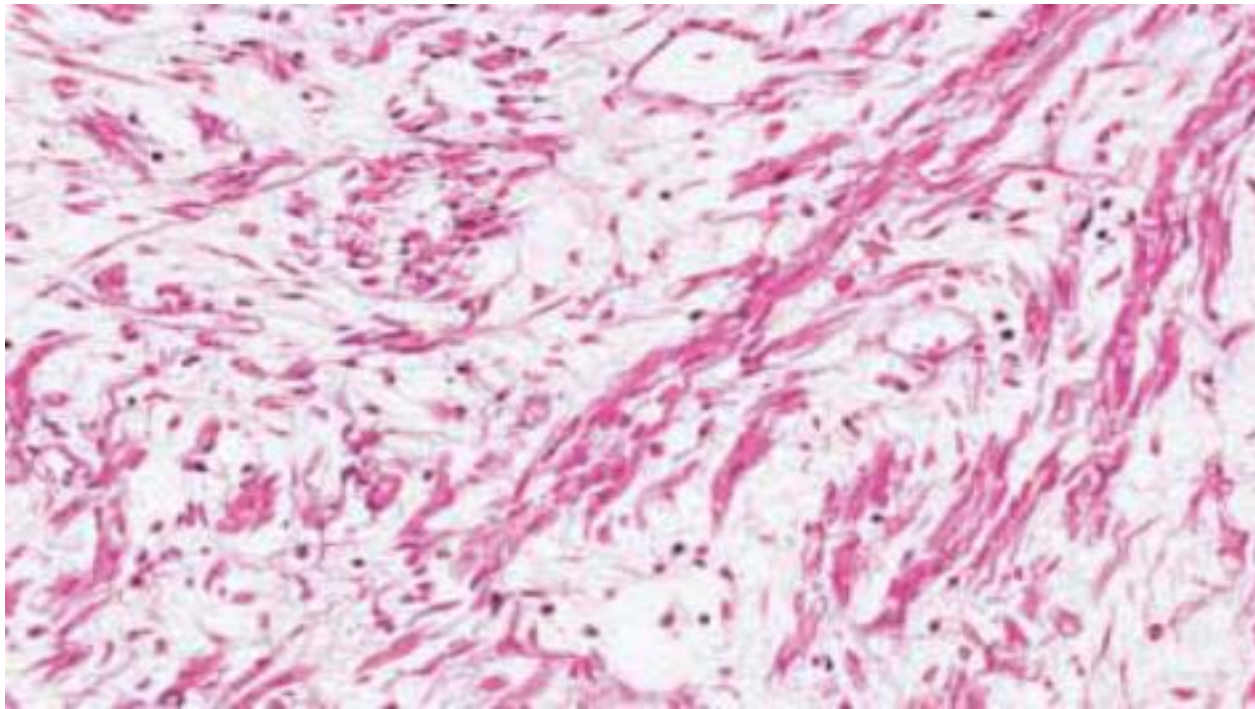


Fig. 6.01 Adult rhabdomyoma. A Well circumscribed mass composed of large polygonal cells with eosinophilic vacuolated cytoplasm and surrounded by normal skeletal muscle. B Higher magnification shows large, polygonal cells with abundant granular and vacuolated cytoplasm.

Fetal Rhabdomyoma

- Fetal rhabdomyoma (F-RM) is a rare benign mesenchymal tumour that exhibits *immature skeletal muscle differentiation* and a predilection for the **head and neck**.



Classic fetal rhabdomyoma. A Tumour is composed of cytologically bland, delicate fetal myotubules. B Primitive spindle cells in a myxoid stroma. C Occasional delicate rhabdomyoblasts display cross striations.

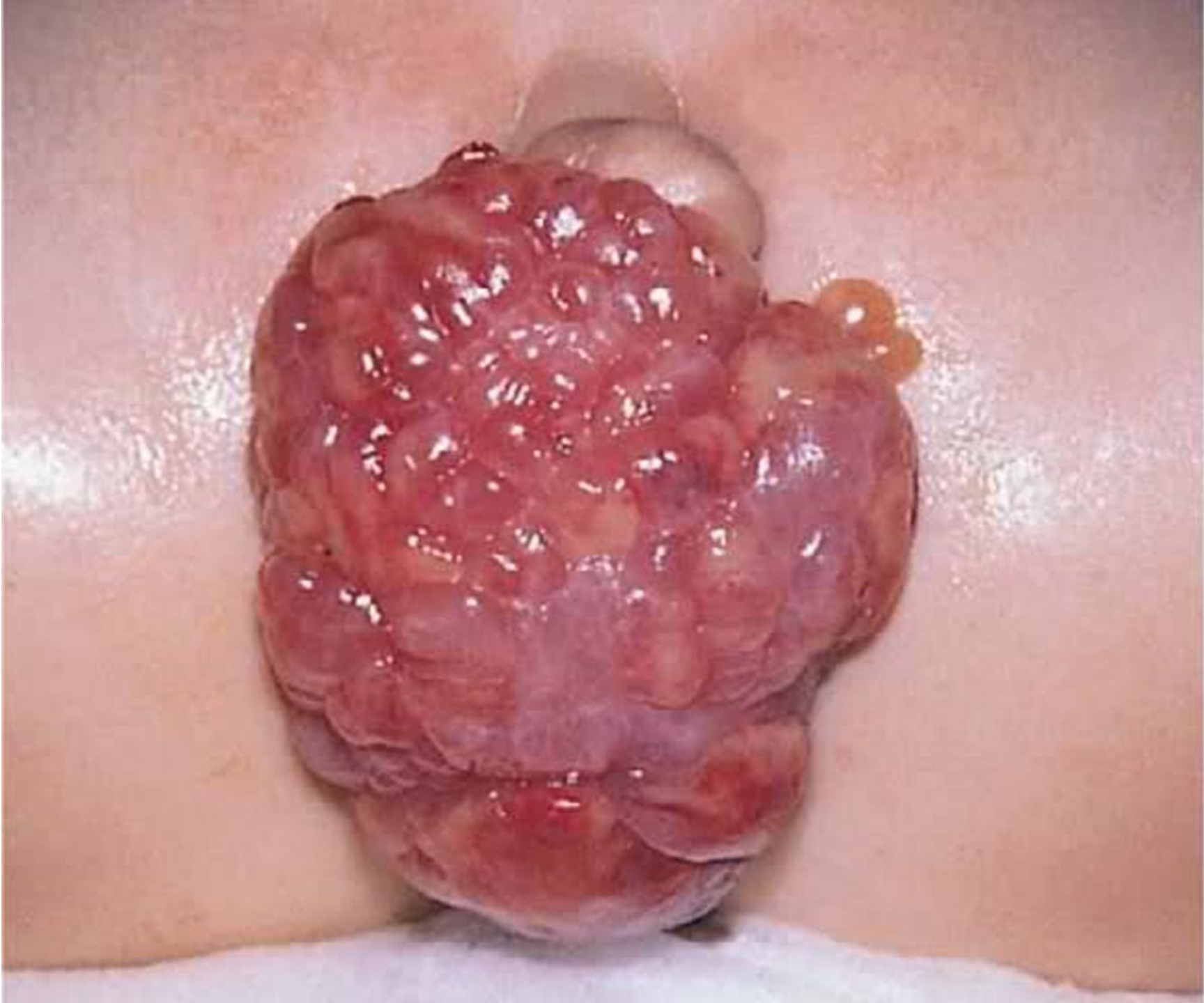
Rhabdomyosarcoma

- **Rhabdomyosarcoma** is the most common soft tissue sarcoma of childhood and adolescence, usually appearing before age 20.
- Interestingly, they occur most commonly in the **head and neck or genitourinary tract**, usually at sites where there is little, if any, skeletal muscle as a normal constituent.

- **Chromosomal translocations** are found in most cases; the more common t(2;13) translocation fuses the PAX3 gene on chromosome 2 with the FKHR gene on chromosome 13.
- PAX3 functions upstream of genes that control skeletal muscle differentiation, and tumor development probably involves dysregulation of muscle differentiation by the chimeric PAX3-FKHR protein.

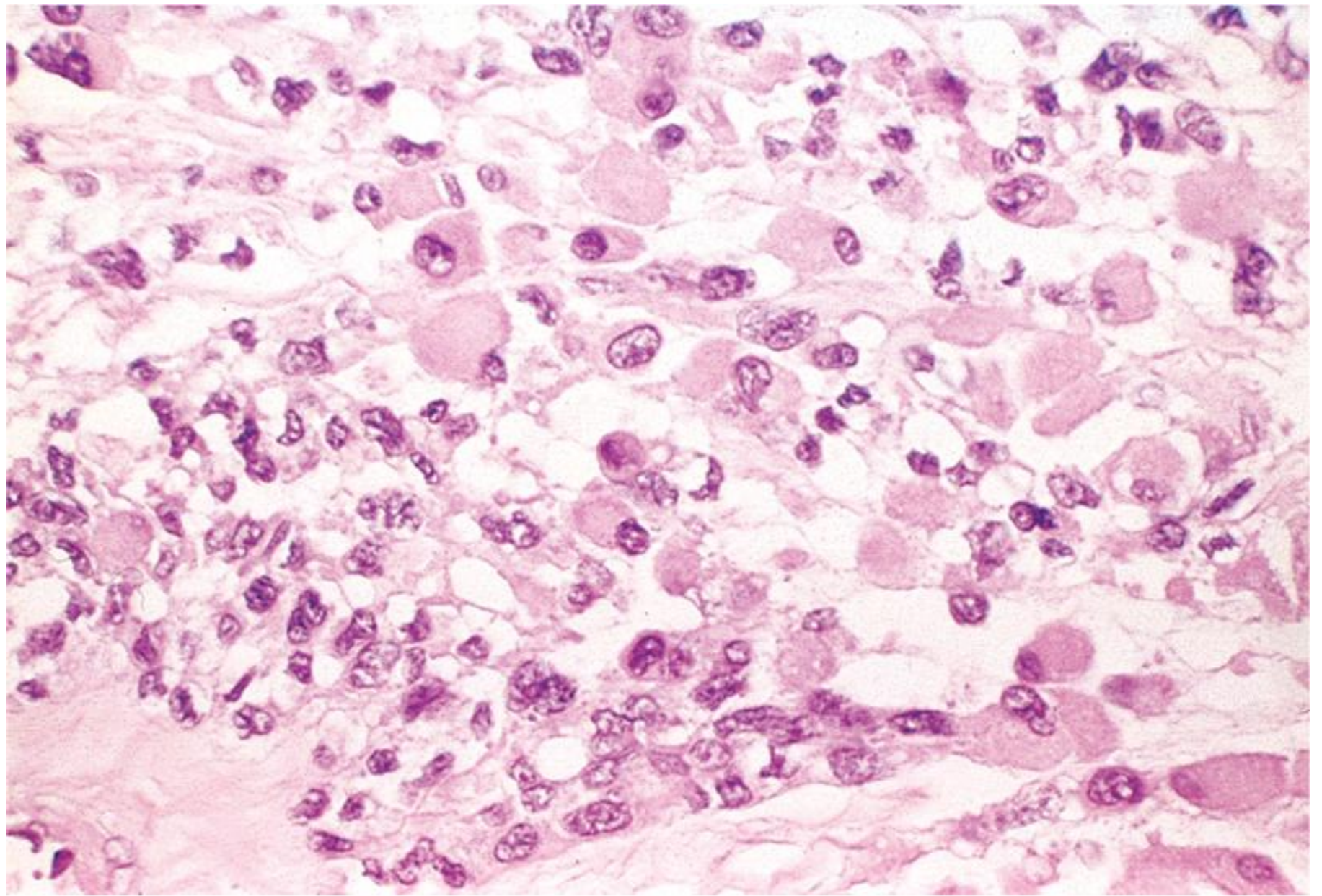
Morphology

- The gross appearance of rhabdomyosarcomas is variable.
- Some tumors, particularly those arising near the mucosal surfaces of the bladder or vagina, can present as soft, gelatinous, grapelike masses, designated **sarcoma botryoides**.
- In other cases they are poorly defined, infiltrating masses.
- Rhabdomyosarcoma is histologically subclassified into the
 - embryonal,
 - alveolar
 - pleomorphic variants.



- The **Rhabdomyoblast** is the **diagnostic cell** in all types; it exhibits granular eosinophilic cytoplasm rich in thick and thin filaments.
- The rhabdomyoblasts may be round or elongated; the latter are known as **tadpole or strap cells** and may contain **cross-striations** visible by light microscopy.

- The diagnosis of rhabdomyosarcoma is based on the demonstration of skeletal muscle differentiation, either in the form of sarcomeres under the electron microscope or by immunohistochemical demonstration of muscle-associated antigens such as desmin and muscle-specific actin.



- Rhabdomyosarcomas are aggressive neoplasms treated with a combination of
 - surgery
 - chemotherapy
 - radiation.

- Location and the histologic variant of the tumor influence survival; **embryonal, pleomorphic, and alveolar variants have progressively worsening prognoses.**
- The malignancy is **curable in almost two-thirds of children**, but *adults do much more poorly.*